

Technical Document #435: Data Engineering - Comprehensive Overview

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Data lineage tracks the origin and movement of data through systems. It is essential for debugging, compliance, and impact analysis.

Data lakes store raw data in its native format. They support structured, semi-structured, and unstructured data at any scale.

Data warehouses store structured data optimized for analytical queries. They use columnar storage and support complex aggregations.

Stream processing handles continuous data streams in real-time. Apache Kafka and Apache Flink are popular stream processing frameworks.

Data governance establishes policies and procedures for managing data assets. It covers data privacy, security, and compliance.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Key Takeaways:

- Data pipelines automate the movement and transformation of data between systems.
- Data lakes store raw data in its native format.
- Schema evolution handles changes in data structure over time.